Kishan Paschapur

reachtokishan@gmail.com | github.com/Kishan-No1

EDUCATION

Irvington High School, Fremont, CA
Junior, Aug. 2024
Diploma, Expected 2028
GPA: 3.83/4.0 UW
GPA: 4.2/4.0 W

De Anza College Early College
Sophomore, June - Aug 2025
Courses (Anticipated + taken):
Intro to Programming in Python, Intro to Statistics, Intro to Microeconomics, Intro to Macroeconomics

RESEARCH EXPERIENCE

A Controlled Analysis of Physics-Informed Augmentation for Powder XRD Classification –Pranav Kulkarni
Accepted, IEEE ONCON 2026
Dec. 2025. - Present

- Built a parallel, resumable pipeline that simulates powder X-ray diffraction patterns from the COD(~490K structures) and rasterizes them for a Swin-T vision transformer.
- Reproduced the published SIMPOD benchmark on the authors' split (82.92% vs. 82.69% reported synthetic top-5 accuracy).
- Designed physics-informed augmentation (peak broadening, background, counting noise, zero shift, texture) that raised top-5 accuracy on 20 real measured patterns from 30% to 55%, with gains in all three repeated runs.
- Ran a leave-one-component-out ablation identifying broadening and background as the main contributors; reported Wilson intervals and exact McNemar tests.

*MIT Beaver Works Summer Institute (**BWSI**)* | *Serious Games & AI (SGAI) Track* | Dr. Rob Seater
Summer 2026

- Co-developed a serious game to study Human-AI trust
- Built the AI pipeline:
  - YOLO detector reading a clear observer view of the road with confidence gated errors to vary its accuracy (~99%/75%/50%)
  - CNN reading player's fogged view
  - DQN agent that plays from both reads
  - All trained on 2,400+ Unity-rendered scenes at three fog levels

ADVANCED COURSEWORK & INDEPENDENT STUDY

AP CSA | AP Physics 1 | APUSH | AP Chem | AP Lang | AP Calc BC *Enrollment – IHS*

Introductory Python Programming *De Anza College course with professor Alex Stoykov*
June. - July. 2025

- A 6 week summer college course completed in the summer of Sophomore Year
- Sharpened OOP skills and overall fluency in Python

Harvard CS50 Introduction to Artificial Intelligence with Python *Self-Study Dec. 2024 - Jan 2025*

- Self-Studied this course over freshman year winter break.
- Built my very own linear regression model with 3 weights to predict housing prices.
- I familiarized myself with many new libraries (Numpy, Pandas, Scikit Learn) and learnt the basics of machine learning techniques.

Stanford CS229 Machine Learning Course *Self-Study Dec. 2025 - Current*

- The most current course I am learning is Andrew Ng's collection of youtube videos.
- Building small projects to familiarize myself with machine learning algorithms.

TECHNICAL EXPERIENCE

Purpose Tech AI Internship *Computer Science Intern + Front End Design Intern* Apr. - June 2024

- Engineered and optimized AI prompts using **Vercel AI**, resolving backend logic issues and improving application reliability
- Improved front-end UX and redesigned automated email templates, increasing user engagement and retention

BAY Python Academy, *Founder, Head of Curriculum and Teaching Department*, Jun. - Aug. 2024

- Initiated a local python academy for advanced elementary schoolers and middle schoolers living in the Bay
- Created and executed an 8-week long course teaching 20+ middle schoolers the basics of python including a capstone project

Extracurriculars + Honors/Awards

ABRSM *Piano Performer/Recitalist*
August, 2018 - Present

- Practicing for ABRSM Grade 8 and recitals
- Composing my own pieces, and studying theory.
- Playing for my local mandir, a religious gathering with over 300+ people weekly

Congressional Award Gold Medal *Submission Under-Review*
Aug. 2024 - Present

- 400+ of Voluntary Public Service, 200+ hours of Personal Development (Piano), 400+ hours physical fitness (Cross Country + Track and Field)
- Led long term service initiatives focused on STEM education, youth mentorship, and community engagement

Innovative Mind for Ethical Technology, *Director of Fremont Branch/Officer of Curriculum and Learning Center,* Fremont, CA (Remote), Aug. - Present. 2025

- Devised social media strategy for a club bringing information and opportunities to kids nationwide in fields related to computer science
- Built Instagram presence; designed posts using Canva.
- Researched topics relating to AI and technology in order to be put on weekly posts
- Led the Fremont division of a nationwide club with members from around the country

LEADERSHIP & SERVICE

Mission Valley Track and Field Club *3x Senior Runner / Captain for Cross Country and Track Team* Aug. 2021 - Aug. 2024

- Initiated team bondings and assisted planning in events that included over 100 elementary and middle schoolers as well as 70+ club alumni
- Collaborated with 5 captains and coaches to spearhead a fundraiser that raised over $35,000 through Snap! Raise.

Fremont Main Library, *Homework Helper + English Buddies,* Fremont, CA,
Aug. 2024 - Present

- Assisted middle schoolers with topics relating to STEM fields and sparked interest in at least 30 children to pursue STEM fields
- So far helped 5 kids conduct research and experiments to submit to local middle school science fairs and advanced to regionals.
- Helped 2 students win the Alameda County Water Districts annual WaterClips contest